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2 May 2026

Joakim Mattbäck
Manager, Technical Expertise, Mechanical & Process
Wärtsilä Energy
Vaasa, Finland

Re: Project Engineer, Process

Dear Joakim Mattbäck,

I am applying for the **Project Engineer, Process** position at Wärtsilä Energy in Vaasa. I am completing my M.Sc. in Sustainable Energy Engineering at KTH Royal Institute of Technology, and this role strongly matches the direction I want to build my career in: process and energy engineering for decarbonised power systems, supported by structured calculations, technical documentation, and international project collaboration.

My background combines energy systems, thermal-process thinking, industrial energy analysis, and hands-on engineering documentation. During my internship at **Alleima AB**, I developed an ISO 50001/EED-aligned energy performance mapping approach for industrial energy systems. This involved defining KPI/EnPI frameworks, baselines, load drivers, normalisation logic, metering readiness, and improvement roadmaps for utilities, process heat, compressed air, and heat recovery. This experience strengthened my interest in process industries and in using data, measurements, and engineering judgement to improve technical systems.

I also bring relevant R&D and testing experience from my master thesis work at **Siemens Energy AB**, where I developed CFD and conjugate heat-transfer models for high-temperature gas-turbine test-rig hardware, commissioned NI-DAQ measurement chains, and documented modelling assumptions, validation limitations, and next-step experimental verification needs. Although this work was not directly PFD/P&ID-based process design, it gave me strong discipline in technical documentation, process understanding, measurement chains, simulation-based analysis, and communicating engineering decisions clearly.

My KTH project work further supports the requirements of this role. I have worked on district-heating dispatch optimisation using Python/PuLP, distribution-grid studies with EV/PV/storage integration, building heating demand forecasting, and energy-system scenario modelling. These projects developed my ability to perform technical calculations, compare alternatives, analyse constraints, and produce structured reports. I am also comfortable with Python, MATLAB basics, Excel-based analysis, and technical writing in English.

What attracts me to Wärtsilä is the opportunity to contribute to engine power plant projects that support the transition toward renewable and flexible energy systems. Wärtsilä's work in balancing power plants, hybrid solutions, energy storage, and optimisation technology fits closely with my interests in sustainable power generation, energy systems modelling, and decarbonisation. I am especially motivated by the chance to grow into process design work involving process flow diagrams, P&IDs, device lists, process descriptions, calculations, and coordination with internal experts and design partners.

I am honest that I am still developing direct professional experience in formal process design documentation such as PFDs and P&IDs. However, I bring a strong energy engineering foundation, industrial process-data experience, simulation and calculation skills, technical documentation discipline, and a strong motivation to learn. I would welcome the opportunity to contribute to Wärtsilä's Process Engineering & Safety team and grow as a project engineer in power plant process design.

Sincerely,
Abhijith Sivaprasadan